



PORTFOLIO

Nsovo Maswanganyi, 224788213

WRDV301, Database Systems

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1. RUBRIC

Class Activities (part of the 60% component of the portfolio mark)

CLASS ACTIVITY	YOUR MARK	MAX MARK	CLASS AVG	COMMENT
01 – 12 FEB*	n	n/a	n/a	Attendance – nothing to include
02 – 19 FEB*	y	n/a	n/a	Attendance – nothing to include
03 – 26 FEB (QUIZ)	6	6	4/6	Include in portfolio
04 – 12 MAR (NORMALISATION)	12	13	11/13	Include in portfolio
05 – 09 APR (MURDER MYSTERY)	10	10	10/10	Include in portfolio
06 – 30 APR (OPTIMISATION)	0	13	8/13	Include in portfolio

* Indicating attendance; no mark allocated

Preparation / assignments (part of the 60% component of the portfolio mark)

ASSIGNMENT	YOUR MARK	MAX MARK	CLASS AVG	Max of one: Either RESUBMIT or FIRST SUBMIT * Indicate with X in relevant column			
				RESUBMIT		FIRST SUBMIT	
				X to resubmit	Mark	X for 1 st submit	Mark
01 – 02 MAR	30	31	25/31				
02 – 16 MAR	51	51	42/51				
03 – 06 APR	45	45	44/45				
04 – 20 APR	26	26	23/26				
05 – 04 MAY	24	24	21/24				

Overview Component	Does the criterion meet the expectations?								Score
	0	No	1	Partially	2	Yes	3	Exceeds	
1. Submitted artefacts	Required sections or artefacts not included		Some sections / artefacts not or not appropriately included		All sections and all artefacts included		All sections and all artefacts appropriately included and documented		
2. Evidence of learning	Little or no evidence of learning		Some evidence of learning, but not sufficient		Adequate evidence of learning		Clear evidence of learning		
3. Evidence of reflection	Little or no evidence of reflection		Some evidence of reflection, but not sufficient		Adequate evidence of reflection		Clear evidence of reflection		
4. Presentation	Instructions not followed; many errors		Most instructions followed		Adequately organised; instructions followed		Well organised, well structured, all instructions followed		

2. CLASS SUBMISSIONS

QUIZ

Science / WRDV301-2026: Database Systems 3.1 (2026) / Class activities / Weak / Strong entities

Weak / Strong entities

Status	Finished
Started	Thursday, 26 February 2026, 8:45 AM
Completed	Thursday, 26 February 2026, 8:46 AM
Duration	1 min 9 secs
Grade	6.00 out of 6.00 (100%)

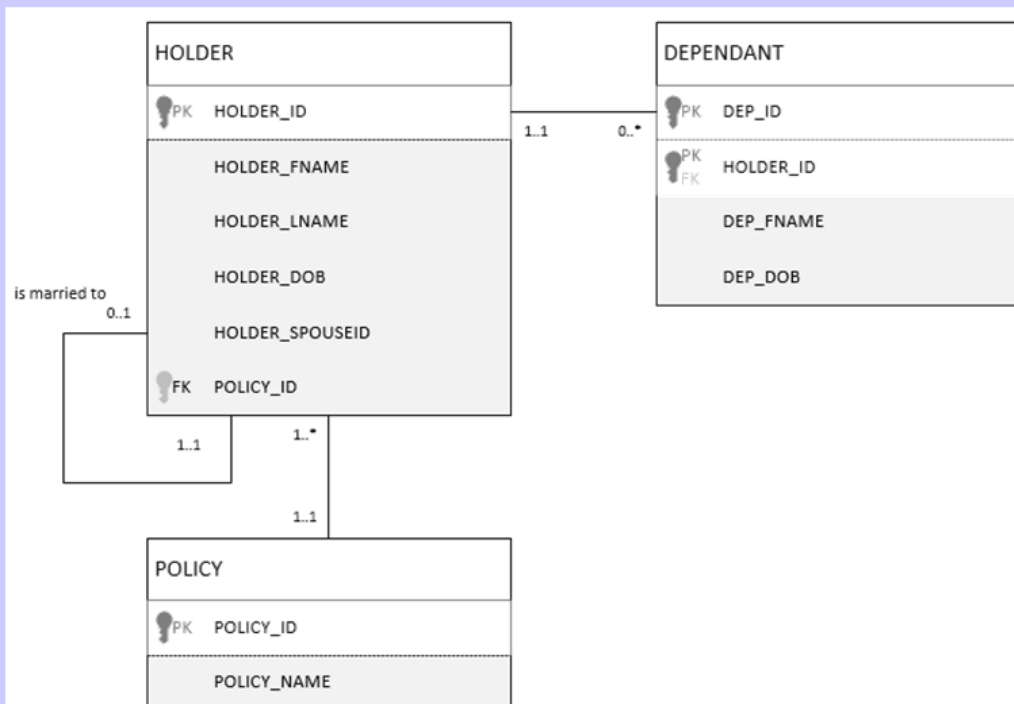
Question 1

Complete

Mark 3.00 out of 3.00

 Flag question

Consider the following diagram. Is DEPENDANT a weak or strong entity?



Select one:

- ☐ Strong - it is existence dependent and its PK is fully derived from the parent entity
- ☐ Strong - it is existence independent and its PK is partially derived from the parent entity
- ☒ Weak - it is existence dependent and its PK is partially derived from the parent entity
- ☐ Strong - it is existence independent and its PK is not derived from the parent entity
- ☐ Weak - it is existence independent and its PK is partially derived from the parent entity
- ☐ Weak - it is existence dependent and its PK is not derived from the parent entity

?

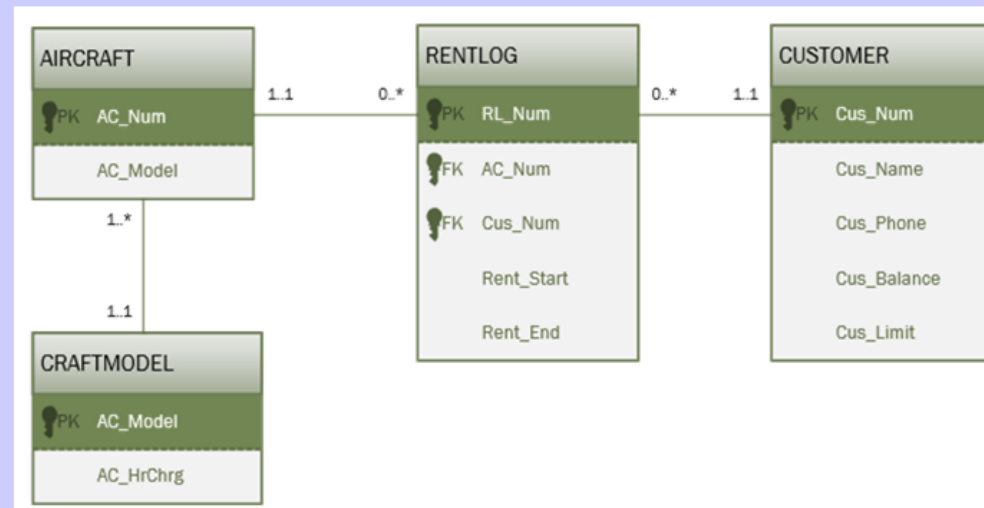
Question 2

Complete

Mark 3.00 out of 3.00

Flag question

Consider the following diagram. Is RENTLOG a weak or strong entity?



Select one:

- ☐ Strong - it is existence dependent and its PK is not derived from the parent entity
- ☐ Weak - it is existence independent and its PK is partially derived from the parent entity
- ☐ Strong - it is existence dependent and its PK is fully derived from the parent entity
- ☐ Strong - it is existence independent and its PK is not derived from the parent entity
- ☒ Weak - it is existence dependent and its natural PK is partially derived from the parent entity
- ☐ Weak - it is existence dependent and its PK is not derived from the parent entity

NORMALIZATION

WRDV301

07:45 Class Exercise

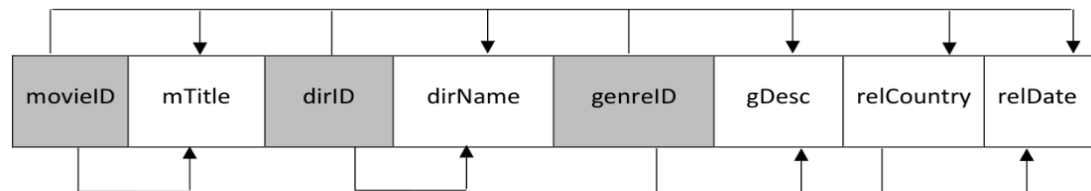
12 March 2026

Group member names:

1. Nsovo Maswanganyi
2. Imbasa Qwabe
- 3.

Scenario

The given dependency diagram shows information about movies, the movie director, the genre of the movie, the country where the movie is released and the release date of the movie.



Instructions:

1. Add the group member names in the header, as indicated.
2. Answer the questions in the table below.
3. Save the file using the surname of the person submitting (only one group member submits).
4. Save Word doc as pdf
5. Submit the pdf in the Normalisation activity for the appropriate session on Funda.

1.	What is the desirable dependency?	1
	Your answer: movieID, dirID, genreID → mTitle, dirName, gDesc, relCountry, relDate	1
2.	List all less-desirable (undesirable) dependencies; and identify the type of less-desirable dependency. Give the dependencies in the appropriate notation, e.g. A → B	3
	Your answer: Partial: movieID → mTitle dirID → dirName genreID → gDesc) ✓✓ Transitive: relCountry → relDate ✓	3
3.	In which NF is this relation? Explain.	1
	Your answer: 1NF. Explanation: The data is in a table format, no repeating groups, the PK is identified, and there are partial dependencies. ✓	1
4.	What is the relational schema for this relation?	1

WRDV301

07:45 Class Exercise

12 March 2026

Group member names:

1. Nsovo Maswanganyi
2. Imbasa Qwabe
- 3.

	Your answer: Movie (movieID, dirID, genreID, mTitle, dirName, gDesc, relCountry, relDate)	1
5.	Normalise this relation into 3NF tables. Use only attributes provided. You need only provide the 3NF relations.	7
	Your answer: Movie (movieID, mTitle) ✓ Director (dirID, dirName) ✓ Genre (genreID, gDesc) ✓ ✓ MovieReleased (movieID, dirID, genreID) Missing an attribute Release (relCountry, relDate) ✓✓	6
	TOTAL	12
		13

MURDER MYSTERY

WRDV301

07:45 Class Exercise

9 April 2026

Group member names:

1. Elbert Joubert
2. Nsovo Maswanganyi
- 3.

Instructions:

1. Add the group member names in the header.
2. Suggestion as to process to follow
 - a. The database design is available on the site – use this to plan your queries
 - b. Start off with what you **know** (what was given)
 - c. Take note of the queries you run, the order in which you run the queries, and the resultant sets returned – this might help you figure out the sequence to get to the answers.
 - d. Do NOT submit this as part of your answer.
3. Answer the questions as explained in the table below.
4. Save the file using the surname of the person submitting (only one group member submits).
5. Save Word doc as pdf
6. Submit the pdf in the Murder mystery activity for the appropriate session on Funda.

1	Who is the murderer?	5
Insert a snippet of your SQL here: Use your knowledge of the database schema and SQL commands to find out who committed the murder. When you think you know the answer, go to the next section. <pre>1 SELECT * 2 FROM person 3 JOIN get_fit_now_member ON person.id = get_fit_now_member.person_id 4 JOIN drivers_license ON person.license_id = drivers_license.id 5 WHERE get_fit_now_member.id LIKE '482%' 6 AND get_fit_now_member.membership_status = 'gold' 7 AND drivers_license.plate_number LIKE '%H42W%' 8 9 10 11 12 13 14</pre> RUN ↴ RESET Check your solution Did you find the killer? <pre>1 INSERT INTO solution VALUES (1, 'Jeremy Bowers'); 2 3 SELECT value FROM solution;</pre> RUN ↴ RESET		

WRDV301

07:45 Class Exercise

9 April 2026

Group member names:

1. Elbert Joubert
2. Nsovo Maswanganyi
- 3.

Insert the message displayed after you've run your SQL code here:

id	name	license_id	address_number	address_street_name	ssn	id	person_id	nam
67318	Jeremy Bowers	423327	530	Washington Pl, Apt 3A	871539279	48Z55	67318	Jer Bo

value

Congrats, you found the murderer! But wait, there's more... If you think you're up for a challenge, try querying the interview transcript of the murderer to find the real villain behind this crime. If you feel especially confident in your SQL skills, try to complete this final step with no more than 2 queries. Use this same INSERT statement with your new suspect to check your answer.

2 Who is the mastermind behind the murder?

5

Insert a snip of your SQL here:

Use your knowledge of the database schema and SQL commands to find out who committed the murder.

When you think you know the answer, go to the next section.

```
1 SELECT *
2 FROM drivers_license
3 JOIN person ON drivers_license.id = person.license_id
4 JOIN facebook_event_checkin ON person.id = facebook_event_checkin.person_id
5 WHERE drivers_license.height BETWEEN 65 AND 67
6 AND drivers_license.hair_color = 'red'
7 AND drivers_license.car_make = 'Tesla'
8 AND drivers_license.car_model = 'Model S'
9 AND facebook_event_checkin.event_name = 'SQL Symphony Concert'
10 AND facebook_event_checkin.date LIKE '201712%'
11 GROUP BY person.name
12 HAVING COUNT(*) = 3
13
14
15
```

RUN ↴

RESET

Check your solution

Did you find the killer?

```
1 INSERT INTO solution VALUES (1, 'Miranda Priestly');
2
3 SELECT value FROM solution;
```

RUN ↴

RESET

WRDV301

07:45 Class Exercise

9 April 2026

Group member names:

1. Elbert Joubert
2. Nsovo Maswanganyi
- 3.

Insert the message displayed after you've run your SQL code here:

id	age	height	eye_color	hair_color	gender	plate_number	car_make	car_model	id	name
202298	68	66	green	red	female	500123	Tesla	Model S	99716	Mirand Priestly

value

Congrats, you found the brains behind the murder! Everyone in SQL City hails you as the greatest SQL detective of all time. break out the champagne!

TOTAL

10

1

0

OPTIMIZATION

No Submission

3. ASSIGNMENTS

ASSIGNMENT 1 DRAFT

Lamola Boitumelo, 229246346
 Maswanganyi Nsovo, 224788213
 Assignment 1 Draft
 Due Date: 23 February 2026

23
31

1.

- MODULE and SESSION
 - There is a one-to-many relationship between a Module and a Session.
 - A MODULE can have many Sessions, but a Session can only belong to one Module.
- SESSION and SCHEDULE
 - The relationship between Session and Schedule is a one-to-many relationship.
 - A Session may be Scheduled zero or many times, but each Schedule must belong to exactly one Session
- STUDENT and SCHEDULE
 - The relationship between a Student and Schedule is a one-to-many relationship.
 - A Student must have at least one Schedule, but each Schedule is associated with exactly one Student.

[6]

2. a)

Table	Primary Key	Foreign Key(s)
STUDENT	stuNum ✓	staffNum ✓
STAFF	staffNum ✓	N/A ✓

[1]

2

b)

Table	Yes / No	Explanation
STUDENT	Yes ✓	Each row has a unique identity. ✓
STAFF	Yes ✓	Each row has a unique identity. ✓

Other pk rule for entity integrity? 2

c)

Table	Yes / No	Explanation
STUDENT	No ✓	The staffNum(FK) 17860 points to a nonexistent entry in the STAFF table ✓
STAFF	No ✓	The table doesn't have foreign keys ✓

3

Lamola Boitumelo, 229246346
 Maswanganyi Nsovo, 224788213
 Assignment 1 Draft
 Due Date: 23 February 2026

3. a) Π stuNum(STUDENT)

Use Insert Equation in Word.

[4]

RESULTANT TABLE

stuNum
22405887
22365169
22478172
22233435
22692868
22345427

2

b) Π stuNum, deptID(STUDENT |X| STAFF)

RESULTANT TABLE

stuNum	deptID
22365169	2119
22478172	1871
22233435	2119
22345427	2034

2

4. a)

[6]

- A Unary relationship exists when an entity is related to itself. An example in this case would be a SalesPerson managing other SalesPersons. In this relationship, one SalesPerson plays the role of a Manager, and the other one plays that of a Subordinate. ✓✓
- A Binary relationship is an association between two entities. This is depicted by a SalesPerson making a Sale. ✓
- A Ternary relationship is a relationship between three entities. An example in this instance is, a SalesPerson makes a Sale, a Product is Sold, and a Customer makes a Purchase. ✓

But here we have 4 entities?

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Maswanganyi Nsovo, 224788213
Assignment 1 Draft
Due Date: 23 February 2026

b) SALE is a weak entity. The existence of a Sale depends on the Product being sold, the Customer making the purchase, the SalesPerson who assisted with the Sale and it also has primary keys that are derived from its parent entities.

3

c) If a new saleID attribute is added to SALE, it would still be a weak entity. For an entity to be weak, the following conditions need to be met:

✓ Correct - argument not completely clear.

- True [
- An entity that is existence-dependent.
 - An entity that has a Primary Key that is partially or totally derived from the parent entity in the relationship.

] Is this still true in this case?

In order to understand the strength of SALE, we first need to outline that it is an associative entity (association class) between the entities SalesPerson and Product. One and only one SalesPerson can sell zero or many products, and one and only one Product can be sold by zero or many SalesPersons (according to the Crow's model provided). This then renders the sale to be completely dependent on those two entities, as well as the Customer who gets involved in a purchase.

1

What are you saying here?

ASSIGNMENT 1 FINAL

Lamola Boitumelo, 229246346
 Maswanganyi Nsovo, 224788213
 Assignment 1
 Due Date: 02 March 2026

30
31

1.

- MODULE and SESSION
 - There is a one-to-many relationship between a Module and a Session.
 - A MODULE can have many Sessions, but a Session can only belong to one Module.
- SESSION and SCHEDULE
 - The relationship between Session and Schedule is a one-to-many relationship.
 - A Session may be Scheduled zero or many times, but each Schedule must belong to exactly one Session
- STUDENT and SCHEDULE
 - The relationship between a Student and Schedule is a one-to-many relationship.
 - A Student must have at least one Schedule, but each Schedule is associated with exactly one Student.

(6)

2. a)

Table	Primary Key	Foreign Key(s)
STUDENT	stuNum	staffNum
STAFF	staffNum	N/A

(8)

2

b)

Table	Yes / No	Explanation
STUDENT	Yes	Each row has a unique identity, and the stuNum attribute is never null. ✓
STAFF	Yes	Each row has a unique identity, and the staffNum attribute is never null. ✓

3

c)

Table	Yes / No	Explanation
STUDENT	No	The staffNum(FK) 17860 points to a nonexistent entry in the STAFF table
STAFF	No	The table doesn't have

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 Maswanganyi Nsovo, 224788213
 Assignment 1
 Due Date: 02 March 2026

		foreign keys
--	--	--------------

3

3. a) $\Pi_{stuNum}(STUDENT)$

RESULTANT TABLE

stuNum
22405887
22365169
22478172
22233435
22692868
22345427

(14)

b) $\Pi_{stuNum, deptID}(STUDENT \bowtie X \bowtie STAFF)$

RESULTANT TABLE

stuNum	deptID
22365169	2119
22478172	1871
22233435	2119
22345427	2034

4. a)

- A Unary relationship exists when an entity is related to itself. An example in this case would be a SalesPerson managing other SalesPersons. In this relationship, one SalesPerson plays the role of a Manager, and the other one plays that of a Subordinate. ✓
- A Binary relationship is an association between two entities. This is depicted by the relationship between Product and ProductCategory. ✓
- A Ternary relationship is a relationship between three entities. An example in this instance is a SalesPerson sells a Product to a Customer. This is a ternary relationship because the associative entity Sale represents a single transaction event that ✓

(12)

Lamola Boitumelo, 229246346
Maswanganyi Nsovo, 224788213
Assignment 1
Due Date: 02 March 2026

simultaneously involves all three parent entities (SalesPerson, Product and Customer).

6

b) SALE is a weak entity. The existence of a Sale depends on the Product being sold, the Customer making the purchase, the SalesPerson who assisted with the Sale, and it also has primary keys that are derived from its parent entities.

3

c) If a new saleID attribute is added to SALE, it would still be a weak entity. This is because Sale is an associative entity that represents a single transaction event that simultaneously involves three parent entities. So, a Sale only makes sense when a Salesperson sells a Product to a Customer. All of these make Sale existence-dependent on SalesPerson, Product and Customer. Another reason is that saleID is added as a surrogate key, which implies that function-wise, the sale entity doesn't change. The key is added only to ease the process of implementing the database.

3

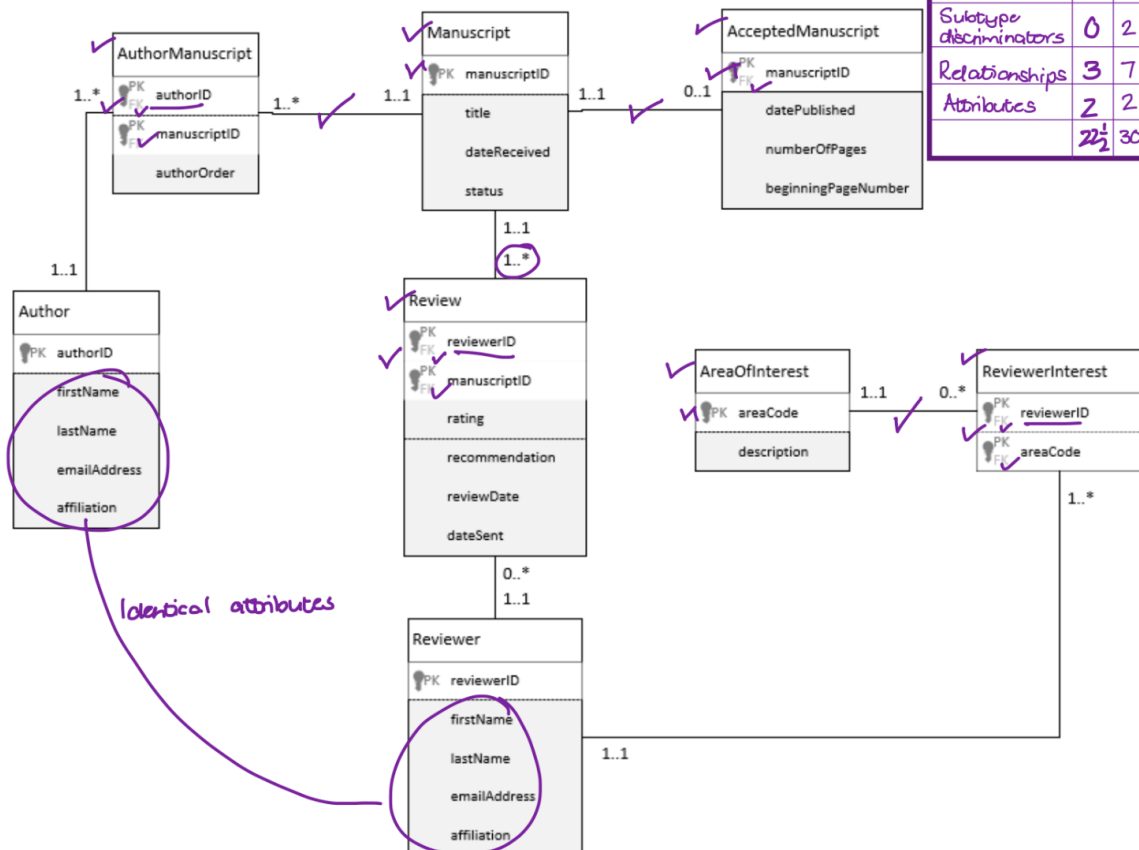
Are the fks mandatory or not? Why?
What is the natural pk?

ASSIGNMENT 2 DRAFT

Lamola Boitumelo, 229246346
 Maswanganyi Nsovo, 224788213
 Assignment 2 Draft
 Due Date: 09 March 2026

Modelling and Normalisation

1. Database design.



Lamola Boitumelo, 229246346
 Maswanganyi Nsovo, 224788213
 Assignment 2 Draft
 Due Date: 09 March 2026

2.

a) CarHire (carReg[pk][fk], cNo[pk][fk], make, cName, cPhone, hireDate, outletNo, outletLoc) ✓ 1

b) **Desirable dependencies:**

- carReg, cNo → make, cName, cPhone, hireDate, outletNo, outletLoc ✓✓

Undesirable dependencies

Transitive dependency:

- outletNo → outletLoc ✓✓

Partial dependency:

- carReg → make, outletNo, outletLoc No ✓

Assumption: they say a car is registered at a specific outlet, meaning if we have a carReg, we can also know where that car is located. Check data given 7

- cNo → cName, cPhone ✓✓

c) Normalisation

- CAR(carReg[pk], make, outletNo[pk])

Assumption: to eliminate the transitive dependency from the CAR table, we eliminated the dependent outletLoc, created another table (OUTLET) and linked both tables by outletNo.

- CUSTOMER(cNo[pk], cName, cPhone) ✓ 3

- CAR HIRED(carReg[pk][fk], cNo[pk][fk], hireDate). Missing attribute

- OUTLET(outletNo[pk], outletLoc) ✓

3. The dependency diagram is in the First Normal Form(1NF), since the NMU data provided is arranged in a **table format**, with **no repeating groups**, as well as an **established composite primary key (PK)** of barCode, empNo. ✓✓

Dependencies are as follows:

Desired dependency: (barCode, empNo → itemDesc, rmNo, bldgNo, bldName, mgrName)

Undesired dependencies

Partial dependencies: - (barCode → itemDesc, rmNo) ✓✓

- (empNo → mgrName) ✓✓

Transitive dependency: (bldgNo → bldName) ✓✓ 7

ASSIGNMENT 2 FINAL

Lamola Boitumelo, 229246346
 Maswanganyi Nsovo, 224788213
 Assignment 2 Final
 Due Date: 16 March 2026

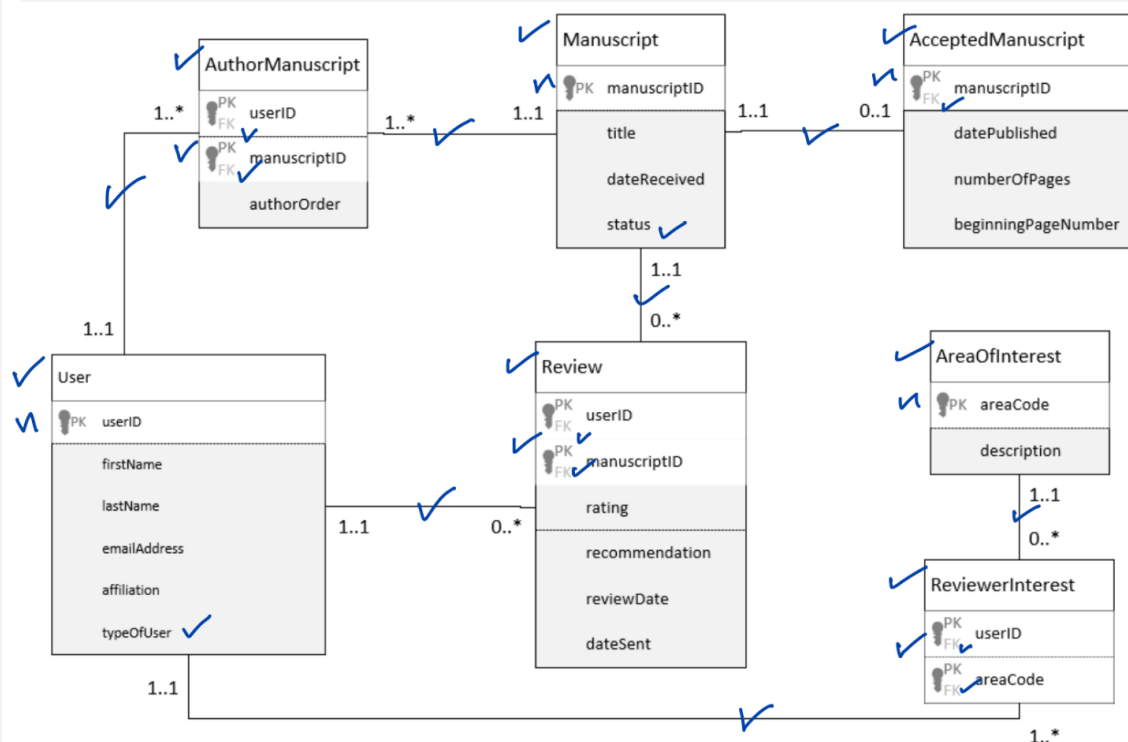
51/51

Modelling and Normalisation

1. Database design.

Updated version 1

	Mark	Max
Entities	7	7
PK	5	5
FK	7	7
Subtype discriminators	2	2
Relationships	7	7
Attributes	2	2
	30	30



Lamola Boitumelo, 229246346
 Maswanganyi Nsovo, 224788213
 Assignment 2 Final
 Due Date: 16 March 2026

2.

a) CarHire (carReg[pk][fk], cNo[pk][fk], make, cName, cPhone, hireDate, outletNo, outletLoc) 1

b) **Desirable dependencies:**

- carReg, cNo → make, cName, cPhone, hireDate, outletNo, outletLoc ✓✓

Undesirable dependencies

Transitive dependency:

- outletNo → outletLoc ✓✓ 8

Partial dependency:

- carReg → make ✓✓

- cNo → cName, cPhone ✓✓

c) Normalisation

- CAR(carReg[pk], make) ✓

- CUSTOMER(cNo[pk], cName, cPhone) ✓

- CAR HIRED(carReg[pk][fk], cNo[pk][fk], hireDate, outletNo) ✓✓ 5

- OUTLET(outletNo[pk], outletLoc) ✓

3. The dependency diagram is in the First Normal Form(1NF), since the NMU data provided is arranged in a **table format**, with **no repeating groups**, as well as an **established composite primary key (PK)** of barCode, empNo.

Dependencies are as follows:

Desired dependency: (barCode, empNo → itemDesc, rmNo, bldgNo, bldName, mgrName)

Undesired dependencies

Partial dependencies: - (barCode → itemDesc, rmNo)

- (empNo → mgrName)

Transitive dependency: (bldgNo → bldName)

ASSIGNMENT 3 DRAFT

Lamola Boitumelo, 229246346
Maswanganyi Nsovo, 224788213
Assignment 2 Draft
Due Date: 23 March 2026

45
45

SQL

1.

`CREATE TABLE EMP_1 (
empNum INTEGER NOT NULL UNIQUE,
empLName VARCHAR(50),
empFName VARCHAR(50),
emplnit NCHAR(10),
empHireDate DATE,
jobCode INTEGER NOT NULL,

PRIMARY KEY(empNum),
FOREIGN KEY(jobCode) REFERENCES JOB
);`

6

2.

a)

1st insertion:

`INSERT INTO EMP_1
VALUES ('101','News','John','G','2022-11-08','502');`

5

2

2nd insertion:

`INSERT INTO EMP_1
VALUES ('102','Senior','David','H','2021-07-12','501');`

Results Messages

	empNum	empLName	empFName	emplnit	empHireDate	jobCode
1	101	News	John	G	2022-11-08	502
2	102	Senior	David	H	2021-07-12	501

Query executed successfully.localhost (17.0 RTM)BOITUMELO_KHANI\lamol ...Assign3DB00:00:00Row: 2, Col: 12 rows

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Assignment 2 Draft
Due Date: 23 March 2026

b)

```
INSERT INTO EMP_1
SELECT
empNum,
empLName,
empFName,
empLname,
empHireDate,
jobCode
FROM Employee
WHERE empNum > 102;
```

3

Results Messages

	empNum	empLName	empFName	emplnit	empHireDate	jobCode
1	101	News	John	G	2022-11-08	502
2	102	Senior	David	H	2021-07-12	501
3	103	Arbough	June	J	2017-12-01	503
4	104	Ramoras	Anne	AC	2009-11-15	501
5	105	Johnson	Alice	A	2014-02-01	502
6	106	Smithfield	William	W	2025-06-22	500
7	107	Alonzo	Maria	MM	2015-10-11	500
8	108	Washington	Ralph	RM	2013-08-22	501
9	109	Smith	Larry	LJ	2018-07-18	501
10	110	Olenko	Gerald	GB	2017-12-11	505
11	111	Wabash	Geoff	GG	2012-04-04	506
12	112	Smithson	Darlene	D	2016-10-23	507
13	113	Joebrood	Delbert	DD	2018-11-15	508
14	114	Jones	Annelise	AS	2016-08-20	508
15	115	Bawangi	Cweka	CX	2013-01-25	501
16	116	Pratt	Gerald	G	2018-03-05	510
17	117	Williamson	Angie	A	2018-06-19	509
18	118	Frommer	James	J	2026-01-04	510
19	119	Smithfield	Will	WBG	2021-06-01	501

Query execu...localhost (17.0 RTM)BOITUMELO_KHANI\lamol ...Assign3DB00:00:00Row: 1, Col: 119 rows

3.

```
UPDATE EMP_1
SET jobCode = 501
WHERE empNum = 107;
```

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Assignment 2 Draft
Due Date: 23 March 2026

Results		Messages				
	empNum	empLName	empFName	empInit	empHireDate	jobCode
1	101	News	John	G	2022-11-08	502
2	102	Senior	David	H	2021-07-12	501
3	103	Arbough	June	J	2017-12-01	503
4	104	Ramoras	Anne	AC	2009-11-15	501
5	105	Johnson	Alice	A	2014-02-01	502
6	106	Smithfield	William	W	2025-06-22	500
7	107	Alonzo	Maria	MM	2015-10-11	501
8	108	Washington	Ralph	RM	2013-08-22	501
9	109	Smith	Larry	LJ	2018-07-18	501
10	110	Olenko	Gerald	GB	2017-12-11	505
11	111	Wabash	Geoff	GG	2012-04-04	506
12	112	Smithson	Darlene	D	2016-10-23	507
13	113	Joenbrood	Delbert	DD	2018-11-15	508
14	114	Jones	Annelise	AS	2016-08-20	508
15	115	Bawangi	Cweka	CX	2013-01-25	501
16	116	Pratt	Gerald	G	2018-03-05	510
17	117	Williamson	Angie	A	2018-06-19	509
18	118	Frommer	James	J	2026-01-04	510
19	119	Smithfield	Will	WBG	2021-06-01	501

Query exec... localhost (17.0 RTM) BOITUMELO_KHANI\lamol ... Assign3DB 00:00:00 Row: 7, Col: 1 19 rows

4.

DELETE FROM EMP_1

WHERE

empFName = 'William' AND

empLName = 'Smithfield' AND

empHireDate = '2025-06-22' AND

jobCode = 500;

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	empNum	empLName	empFName	empInit	empHireDate	jobCode
1	101	News	John	G	2022-11-08	502
2	102	Senior	David	H	2021-07-12	501
3	103	Arbough	June	J	2017-12-01	503
4	104	Ramoras	Anne	AC	2009-11-15	501
5	105	Johnson	Alice	A	2014-02-01	502
6	107	Alonzo	Maria	MM	2015-10-11	501
7	108	Washington	Ralph	RM	2013-08-22	501
8	109	Smith	Larry	LJ	2018-07-18	501
9	110	Olenko	Gerald	GB	2017-12-11	505
10	111	Wabash	Geoff	GG	2012-04-04	506
11	112	Smithson	Darlene	D	2016-10-23	507
12	113	Joenbrood	Delbert	DD	2018-11-15	508
13	114	Jones	Annelise	AS	2016-08-20	508
14	115	Bawangi	Cweka	CX	2013-01-25	501
15	116	Pratt	Gerald	G	2018-03-05	510
16	117	Williamson	Angie	A	2018-06-19	509
17	118	Frommer	James	J	2026-01-04	510
18	119	Smithfield	Will	WBG	2021-06-01	501

Query executed on: localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 5, Col: 1 | 18 rows

5.

```
SELECT *  
INTO EMP_2  
FROM EMP_1;
```

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The screenshot shows a SQL Server Enterprise Manager interface. The top pane displays a query window with the following SQL code:

```
1 SELECT TOP (1000) [empNum]
2     , [empLName]
3     , [empFName]
4     , [empInit]
5     , [empHireDate]
6     , [jobCode]
7 FROM [Assign3DB].[dbo].[EMP_2]
8
```

The bottom pane shows the results of the query, displaying 18 rows of data. The columns are: empNum, empLName, empFName, empInit, empHireDate, and jobCode.

	empNum	empLName	empFName	empInit	empHireDate	jobCode
1	101	News	John	G	2022-11-08	502
2	102	Senior	David	H	2021-07-12	501
3	103	Arbough	June	J	2017-12-01	503
4	104	Ramoras	Anne	AC	2009-11-15	501
5	105	Johnson	Alice	A	2014-02-01	502
6	107	Alonzo	Maria	MM	2015-10-11	501
7	108	Washington	Ralph	RM	2013-08-22	501
8	109	Smith	Larry	LJ	2018-07-18	501
9	110	Olenko	Gerald	GB	2017-12-11	505
10	111	Wabash	Geoff	GG	2012-04-04	506
11	112	Smithson	Darlene	D	2016-10-23	507
12	113	Joenbrood	Delbert	DD	2018-11-15	508
13	114	Jones	Annelise	AS	2016-08-20	508
14	115	Bawangi	Cweka	CX	2013-01-25	501
15	116	Pratt	Gerald	G	2018-03-05	510
16	117	Williamson	Angie	A	2018-06-19	509
17	118	Frommer	James	J	2026-01-04	510
18	119	Smithfield	Will	WBG	2021-06-01	501

The status bar at the bottom indicates: Query execute... localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 1, Col: 1 | 18 rows

6.

ALTER TABLE EMP_2
ADD empPerc DECIMAL;

ALTER TABLE EMP_2
ADD projNum INT;

Can do single ALTER.... Add

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Results		Messages						
	empNum	empLName	empFName	emplnit	empHireDate	jobCode	empPerc	projNum
1	101	News	John	G	2022-11-08	502	NULL	NULL
2	102	Senior	David	H	2021-07-12	501	NULL	NULL
3	103	Arbough	June	J	2017-12-01	503	NULL	NULL
4	104	Ramoras	Anne	AC	2009-11-15	501	NULL	NULL
5	105	Johnson	Alice	A	2014-02-01	502	NULL	NULL
6	107	Alonzo	Maria	MM	2015-10-11	501	NULL	NULL
7	108	Washington	Ralph	RM	2013-08-22	501	NULL	NULL
8	109	Smith	Larry	LJ	2018-07-18	501	NULL	NULL
9	110	Olenko	Gerald	GB	2017-12-11	505	NULL	NULL
10	111	Wabash	Geoff	GG	2012-04-04	506	NULL	NULL
11	112	Smithson	Darlene	D	2016-10-23	507	NULL	NULL
12	113	Joenbrood	Delbert	DD	2018-11-15	508	NULL	NULL
13	114	Jones	Annelise	AS	2016-08-20	508	NULL	NULL
14	115	Bawangi	Cweka	CX	2013-01-25	501	NULL	NULL
15	116	Pratt	Gerald	G	2018-03-05	510	NULL	NULL
16	117	Williamson	Angie	A	2018-06-19	509	NULL	NULL
17	118	Frommer	James	J	2026-01-04	510	NULL	NULL
18	119	Smithfield	Will	WBG	2021-06-01	501	NULL	NULL

✓ Quer... localhost (17.0 RTM) BOITUMELO_KHANI\lamol ... Assign3DB 00:00:00 Row: 1, Col: 1 18 rows

7.

a) **UPDATE** EMP_2
SET empPerc = 4
WHERE empNum = 103;

Results		Messages						
	empNum	empLName	empFName	emplnit	empHireDate	jobCode	empPerc	projNum
1	101	News	John	G	2022-11-08	502	NULL	NULL
2	102	Senior	David	H	2021-07-12	501	NULL	NULL
3	103	Arbough	June	J	2017-12-01	503	4	NULL
4	104	Ramoras	Anne	AC	2009-11-15	501	NULL	NULL
5	105	Johnson	Alice	A	2014-02-01	502	NULL	NULL
6	107	Alonzo	Maria	MM	2015-10-11	501	NULL	NULL
7	108	Washington	Ralph	RM	2013-08-22	501	NULL	NULL
8	109	Smith	Larry	LJ	2018-07-18	501	NULL	NULL

✓ Quer... localhost (17.0 RTM) BOITUMELO_KHANI\lamol ... Assign3DB 00:00:00 Row: 3, Col: 7 18 rows

b) **UPDATE** EMP_2
SET empPerc = 5
WHERE empNum = 101
 OR empNum = 105
 OR empNum = 107;

3
 } There is a more optimal way

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Results

Messages

	empNum	empLName	empFName	empLnit	empHireDate	jobCode	empPerc	projNum
1	101	News	John	G	2022-11-08	502	5	NULL
2	102	Senior	David	H	2021-07-12	501	NULL	NULL
3	103	Arbough	June	J	2017-12-01	503	4	NULL
4	104	Ramoras	Anne	AC	2009-11-15	501	NULL	NULL
5	105	Johnson	Alice	A	2014-02-01	502	5	NULL
6	107	Alonzo	Maria	MM	2015-10-11	501	5	NULL
7	108	Washington	Ralph	RM	2013-08-22	501	NULL	NULL
8	109	Smith	Larry	LJ	2018-07-18	501	NULL	NULL

✓ Quer...

localhost (17.0 RTM)

BOITUMELO_KHANI\lamol ...

Assign3DB

00:00:00

Row: 1, Col: 1

18 rows

- c) **UPDATE** EMP_2
SET empPerc = 10
WHERE empPerc IS NULL;

2

Results		Messages						
	empNum	empLName	empFName	empLnit	empHireDate	jobCode	empPerc	projNum
1	101	News	John	G	2022-11-08	502	5	NULL
2	102	Senior	David	H	2021-07-12	501	10	NULL
3	103	Arbough	June	J	2017-12-01	503	4	NULL
4	104	Ramoras	Anne	AC	2009-11-15	501	10	NULL
5	105	Johnson	Alice	A	2014-02-01	502	5	NULL
6	107	Alonzo	Maria	MM	2015-10-11	501	5	NULL
7	108	Washington	Ralph	RM	2013-08-22	501	10	NULL
8	109	Smith	Larry	LJ	2018-07-18	501	10	NULL
9	110	Olenko	Gerald	GB	2017-12-11	505	10	NULL
10	111	Wabash	Geoff	GG	2012-04-04	506	10	NULL
11	112	Smithson	Darlene	D	2016-10-23	507	10	NULL
12	113	Joenbrood	Delbert	DD	2018-11-15	508	10	NULL
13	114	Jones	Annelise	AS	2016-08-20	508	10	NULL
14	115	Bawangi	Cweka	CX	2013-01-25	501	10	NULL
15	116	Pratt	Gerald	G	2018-03-05	510	10	NULL
16	117	Williamson	Angie	A	2018-06-19	509	10	NULL
17	118	Frommer	James	J	2026-01-04	510	10	NULL
18	119	Smithfield	Will	WBG	2021-06-01	501	10	NULL

✓ Quer...

🔒 localhost (17.0 RTM)

BOITUMELO_KHANI\lamol ...

Assign3DB

00:00:00

Row: 1, Col: 1

18 rows

- d) **UPDATE** EMP_2
SET projNum = 25
WHERE jobCode >= 502;

2

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Results		Messages						
	empNum	empLName	empFName	emplnit	empHireDate	jobCode	empPerc	projNum
1	101	News	John	G	2022-11-08	502	5	25
2	102	Senior	David	H	2021-07-12	501	10	NULL
3	103	Arbough	June	J	2017-12-01	503	4	25
4	104	Ramoras	Anne	AC	2009-11-15	501	10	NULL
5	105	Johnson	Alice	A	2014-02-01	502	5	25
6	107	Alonzo	Maria	MM	2015-10-11	501	5	NULL
7	108	Washington	Ralph	RM	2013-08-22	501	10	NULL
8	109	Smith	Larry	LJ	2018-07-18	501	10	NULL
9	110	Olenko	Gerald	GB	2017-12-11	505	10	25
10	111	Wabash	Geoff	GG	2012-04-04	506	10	25
11	112	Smithson	Darlene	D	2016-10-23	507	10	25
12	113	Joenbrood	Delbert	DD	2018-11-15	508	10	25
13	114	Jones	Annelise	AS	2016-08-20	508	10	25
14	115	Bawangi	Cweka	CX	2013-01-25	501	10	NULL
15	116	Pratt	Gerald	G	2018-03-05	510	10	25
16	117	Williamson	Angie	A	2018-06-19	509	10	25
17	118	Frommer	James	J	2026-01-04	510	10	25
18	119	Smithfield	Will	WBG	2021-06-01	501	10	NULL

✓ Quer...

localhost (17.0 RTM)

BOITUMELO_KHANI\lamol ...

Assign3DB

00:00:00

Row: 1, Col: 1

18 rows

e) **UPDATE** EMP_2
SET projNum = 14
WHERE empHireDate < '2021-07-10' 3
AND jobCode <= 502;

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Results

Messages

	empNum	empLName	empFName	emplnit	empHireDate	jobCode	empPerc	projNum
1	101	News	John	G	2022-11-08	502	5	25
2	102	Senior	David	H	2021-07-12	501	10	NULL
3	103	Arbough	June	J	2017-12-01	503	4	25
4	104	Ramoras	Anne	AC	2009-11-15	501	10	14
5	105	Johnson	Alice	A	2014-02-01	502	5	14
6	107	Alonzo	Maria	MM	2015-10-11	501	5	14
7	108	Washington	Ralph	RM	2013-08-22	501	10	14
8	109	Smith	Larry	LJ	2018-07-18	501	10	14
9	110	Olenko	Gerald	GB	2017-12-11	505	10	25
10	111	Wabash	Geoff	GG	2012-04-04	506	10	25
11	112	Smithson	Darlene	D	2016-10-23	507	10	25
12	113	Joebrood	Delbert	DD	2018-11-15	508	10	25
13	114	Jones	Annelise	AS	2016-08-20	508	10	25
14	115	Bawangi	Cwaka	CX	2013-01-25	501	10	14
15	116	Pratt	Gerald	G	2018-03-05	510	10	25
16	117	Williamson	Angie	A	2018-06-19	509	10	25
17	118	Frommer	James	J	2026-01-04	510	10	25
18	119	Smithfield	Will	WBG	2021-06-01	501	10	14

✓ Quer...

localhost (17.0 RTM)

BOITUMELO_KHANI\lamol ...

Assign3DB

00:00:00

Row: 1, Col: 1

18 rows

f) `SELECT empNum, empLName, empFName
 FROM EMP_2
 WHERE empLName LIKE 'Smith%'
 ORDER BY empLName DESC;`

3

Results		Messages	
	empNum	empLName	empFName
1	112	Smithson	Darlene
2	119	Smithfield	Will
3	109	Smith	Larry

✓ Qu... | localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 1, Col: 1 | 3 rows

8.
`SELECT Employee.empNum,
 empLName,
 SUM(assignHrs) AS sumHRS,
 SUM(assignChrg) AS sumCHRG
 FROM Employee JOIN Assignment ON Employee.empNum = Assignment.empNum
 GROUP BY Employee.empNum, empLName;`

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Results		Messages		
	empNum	empLName	sumHRS	sumCHRG
1	101	News	3	387.50
2	103	Arbough	20	1664.65
3	104	Ramoras	12	1218.70
4	105	Johnson	9	931.00
5	108	Washington	8	840.15
6	113	Joebrood	4	192.85
7	115	Bawangi	13	1276.75
8	117	Williamson	18	649.54

BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 1, Col: 1 | 8 rows

9.
`SELECT projNum,
SUM (assignHrs) AS sumHRS,
SUM (assignChrg) AS sumCHRG
FROM Assignment
GROUP BY projNum
ORDER BY projNum DESC;`

Results		Messages		
	projNum	sumHRS	sumCHRG	
1	25	19	1668.16	
2	22	28	2593.16	
3	18	24	1544.80	
4	15	20	1806.52	

localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 1, Col: 1 | 4 rows

ASSIGNMENT 3 FINAL

Same as draft

ASSIGNMENT 4 DRAFT

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Maswanganyi Nsovo, 224788213
Assignment 4 Draft
Due Date: 13 April 2026

17½
26

Database Development and Design

1.

a) 3 Ways of implementing the conceptual model

Option 1:

Creating one table for the superclass

This table would have:

- Its attributes and all the attributes from both the subclasses
- A subtype discriminator (empType) to distinguish between the two types of employees ✓
- In an instance where the Employee is a Doctor, the attributes of the Nurse will be null. ✓
- Attributes: empNo[pk], empFname, empLname, empDoB, empType, position, shiftHours ✓

Option 2:

Implementing all three tables and having subclasses associated with the super class through appropriate 1-1 relationships.

- Each subtype would have its unique attributes, as well as a foreign key from the parent class, which would be shown as [pk][fk] in the design. ✓
- Employee: empNo[pk], empFname, empLname, empDoB
- Doctor: empNo[pk][fk], position ✓
- Nurse: empNo[pk][fk], shiftHours

Subtype discriminator?

Option 3:

Implementing only the subtype classes

- Each subtype must have all the attributes of the supertype class and its unique attributes. ✓
- The primary key will remain the same as the one in the supertype class
- Doctor: empNo[pk], empFname, empLname, empDoB, position ✓
- Nurse: empNo[pk], empFname, empLname, empDoB, shiftHours. ✓

8

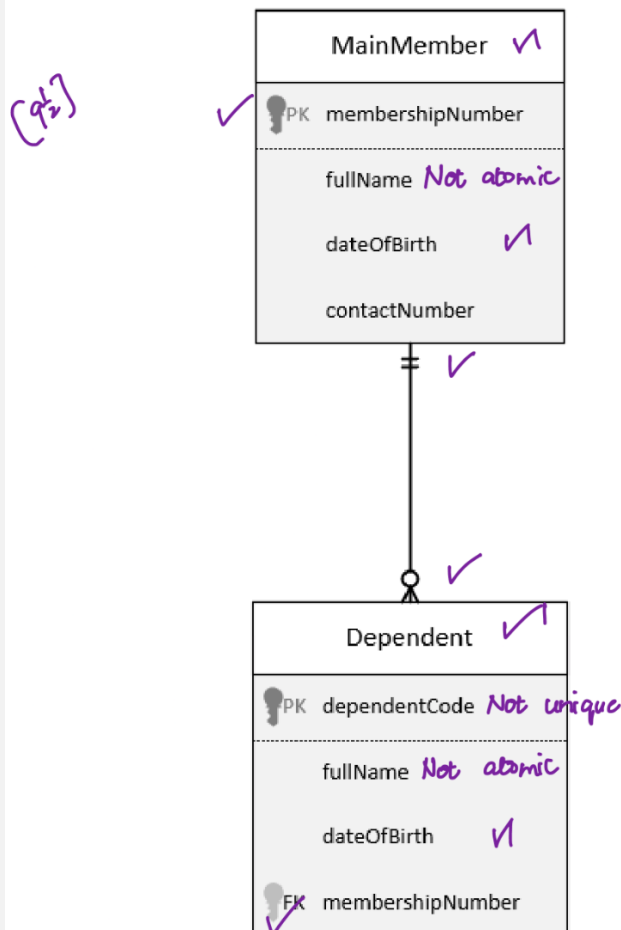
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- b) For the current situation, implementing all tables is best (Our option 2 above). Doctor and Nurse have at least one unique attribute, so to avoid nulls when **either(OR)** employee is instantiated, each class should have its own implementation. Both subclasses would now be associated with the parent class through a 1-1 relationship. This option will also reduce redundancy and update anomalies by storing shared employee data once in the Employee table and subtype tables storing only what is unique to them.

Is the supertype abstract in the scenario, or not?

2.

a)



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b) Relational Schema

- MAINMEMBER (membershipNumber[pk], fullName, dateOfBirth, contactNumber) ✓✓
Not atomic
- DEPENDENT (dependentCode[pk], fullName, dateOfBirth, membershipNumber[fk]) ✓✓ $3\frac{1}{2}$
Only fk?

ASSIGNMENT 4 FINAL

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Assignment 4
Due Date: 20 April 2026

26
26

Database Development and Design

1.

a) 3 Ways of implementing the conceptual model

Option 1:

Creating one table for the superclass

This table would have:

- Its attributes and all the attributes from both the subclasses
- A subtype discriminator (empType) to distinguish between the two types of employees
- In an instance where the Employee is a Doctor, the attributes of the Nurse will be null.
- Attributes: empNo[pk], empFname, empLname, empDoB, empType, position, shiftHours

Option 2:

Implementing all three tables and having subclasses associated with the super class through appropriate 1-1 relationships.

- Each subtype would have unique attributes and a foreign key from the parent class, which would be shown as [pk][fk] in the design.
- A subtype discriminator (empType) to distinguish between the two types of employees ✓
- Employee: empNo[pk], empFname, empLname, empDoB, empType
- Doctor: empNo[pk][fk], position
- Nurse: empNo[pk][fk], shiftHours

Option 3:

Implementing only the subtype classes

- Each subtype must have all the attributes of the supertype class and its unique attributes.
- The primary key will remain the same as the one in the supertype class
- Doctor: empNo[pk], empFname, empLname, empDoB, position
- Nurse: empNo[pk], empFname, empLname, empDoB, shiftHours. 9

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b)

For the current situation, implementing only the subtype tables is best (Our option 3 above).

In database modelling, a **mandatory completeness constraint** implies that every instance of the supertype (EMPLOYEE) **must** belong to at least one of the subtypes (DOCTOR or NURSE). Which also implies the EMPLOYEE class is **abstract**.

This means we would only have 2 classes, DOCTOR and NURSE, and would have all attributes from EMPLOYEE as well as their unique attributes.

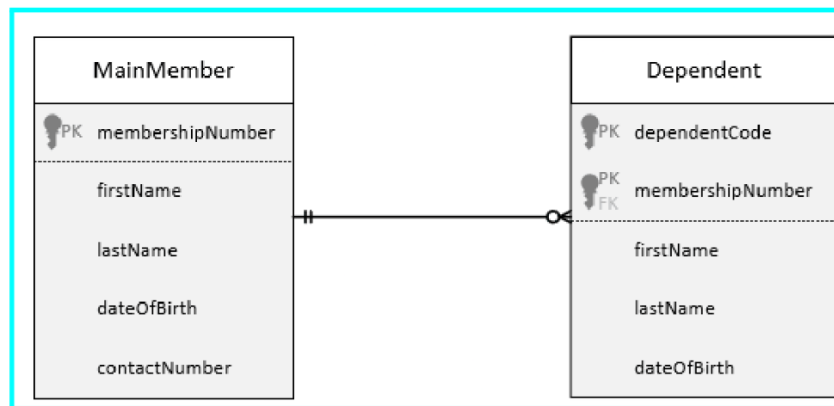
DOCTOR (empNo[pk], empFname, empLname, empDoB, position)
 NURSE (empNo[pk], empFname, empLname, empDoB, shiftHours)

4

2.

a)

[13]



8

b) Relational Schema

- MAINMEMBER (membershipNumber[pk], firstName, lastName, dateOfBirth, contactNumber)
- DEPENDENT (dependentCode[pk], membershipNumber[pk][fk], firstName, lastName, dateOfBirth)

W

5

ASSIGNMENT 5 DRAFT

Lamola Boitumelo, 229246346
 Maswanganyi Nsovo, 224788213
 Assignment 5 Draft
 Due Date: 27 April 2026

19
 24

1. Wait/Die Scheme

Timestamps

T1=101, T2=102, T3=103, T4=104, T5=105.

T3 R(X)	T3 requests and is granted a read-lock on X ✓
T5 R(X)	T5 requests and is granted a read-lock on X ✓
T1 W(X)	T1 requests an exclusive lock on X, T3 and T5 hold a read-lock on X, T1 is older than both T3 and T5, T1 waits for T3 and T5 to complete ✓
T2 W(Y)	T2 requests and is granted an exclusive lock on Y ✓
T3 R(Y)	T3 requests a read-lock on Y, T2 holds an exclusive lock on Y, T3 is younger than T2, T3 dies rolls back and is rescheduled with the original timestamp ✓
T5 W(Z)	T5 requests and is granted an exclusive lock on Z ✓
T5 COMMITS	T5 releases all locks <i>Which locks? And what else happens?</i>
T2 R(X)	T2 requests and is granted a read-lock on X ✓

[7]

2. Why was T1 aborted?

(i) "Uncommitted data" problem.

(ii) Timestamp = 8 ✓

(iii) Data that is not committed is a problem because, in this case, Transaction 2 wants access to item C while Transaction 1 is not done with it, i.e. has not committed yet. This means transaction 1 in Time 8 ~~gained~~ access to a temporary data item, and this causes integrity issues whenever there's a rollback

[5]

At what point was this created / caused?

The transaction property that is violated is the **isolation property**, which states that data cannot be used by a 2nd transaction until the 1st transaction is done with the data(commits).

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Maswanganyi Nsovo, 224788213
Assignment 5 Draft
Due Date: 27 April 2026

3.

[1] **FROM** vehicle
JOIN constructor **ON** vehicle.conCode = constructor.conCode ✓
JOIN service **ON** vehicle.vin = service.vin ✓
WHERE conName = 'Ferrari' ✓
AND serviceType LIKE '%oil%' ✓

Explanation:

The SQL query is now optimised because we are using "JOIN" statements instead of old-style joins. Using JOIN also optimises the filtering process in the WHERE clause. The equality operation is now the first to be performed, which narrows the dataset to only 2 records (since there are 2 Ferraris), and other operations, such as LIKE, can be applied to a smaller dataset, unlike before. ✓

ASSIGNMENT 5 FINAL

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 Maswanganyi Nsovo, 224788213
 Assignment 5
 Due Date: 04 May 2026

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 24

1. Wait/Die Scheme

Timestamps

T1=101, T2=102, T3=103, T4=104, T5=105.

T3 R(X)	T3 requests and is granted a read-lock on X
T5 R(X)	T5 requests and is granted a read-lock on X
T1 W(X)	T1 requests an exclusive lock on X, T3 and T5 hold a read-lock on X, T1 is older than both T3 and T5, T1 waits for T3 and T5 to complete
T2 W(Y)	T2 requests and is granted an exclusive lock on Y
T3 R(Y)	T3 requests a read-lock on Y, T2 holds an exclusive lock on Y, T3 is younger than T2, T3 dies/ rolls back and is rescheduled with the original timestamp
T5 W(Z)	T5 requests and is granted an exclusive lock on Z
T5 COMMITS	T5 releases read-lock on X and exclusive lock on Z. T1 is granted exclusive lock on X.
T2 R(X)	T2 requests read-lock on X, T1 holds exclusive lock on X, T2 is younger than T1, T2 dies and is rescheduled with the original timestamp.

[11]

2. Why was T1 aborted?

(i) "Uncommitted data" problem.

(ii) Timestamp = 8

(iii) Data that is not committed is a problem because, in this case, Transaction 2 wants access to item C while Transaction 1 is not done with it, i.e. has not committed yet. This means transaction 1 in Time 8 gained access to a temporary data item that was created (Written) in Time 6, and this causes integrity issues whenever there's a

[6]

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Maswanganyi Nsovo, 224788213
Assignment 5
Due Date: 04 May 2026

rollback.

The transaction property that is violated is the **isolation property**, which states that data cannot be used by a 2nd transaction until the 1st transaction is done with the data(commits).

3.

FROM vehicle
JOIN constructor **ON** vehicle.conCode = constructor.conCode
JOIN **service ON** vehicle.vin = **service.vin**
WHERE conName = 'Ferrari'
AND serviceType **LIKE** '%oil%'

Explanation:

The SQL query is now optimised because we are using "JOIN" statements instead of old-style joins. Using JOIN also optimises the filtering process in the WHERE clause. The equality operation is now the first to be performed, which narrows the dataset to only 2 records (since there are 2 Ferraris), and other operations, such as LIKE, can be applied to a smaller dataset, unlike before.

4. REFLECTION

Databases are a critical concept in our daily lives. To think that everything we interact with on the internet is stored on some database is something that truly blew my mind. It made me realize. It made me realize that behind every simple search, every online purchase, and every social media post, there is a carefully designed system behind the scenes.

I have to say that I've been very fortunate to enroll in this module, and being able to draw insights from Dr Barnard on how databases work will be one of the highlights that will stay with me for the rest of my life. Her explanations often went beyond the textbook, showing us how these concepts apply in real-world systems, and that perspective will remain valuable to me.

Below I will go over some aspects that will serve as proof that I have fully participated in this module.

THE MOST VALUABLE CONCEPT

Although there are two concepts that stand out for me (Database Design and Distributed Database Management), in this reflection, I will be focusing on Database Design.

Database design act as a foundation of all the databases in the world, and without that foundation, data interaction would most likely be inefficient and unreliable. It is almost like a blueprint of a building; without a solid plan, the structure may not stand firmly, and everything built on top may eventually collapse.

Put simply, database design is the process of planning, structuring, and organizing data according to a specific database model to ensure efficient storage, quick retrieval, and long-term data integrity. What struck me most was how much thought goes into this stage before any queries are written. Sub concepts such as normalization, indexing, tuning, and defining relationships between tables helped me realize that database design is much more than just storing data, but also about anticipating future needs and making sure that the system can scale without crumbling.

Understanding Database design helped me realize that performance is not solely dependent on algorithms alone, but that a well-designed database enables and supports those algorithms, allowing them to work at their best. A poorly structured database can make even the most advanced algorithms struggle, while a well-structured one can make ordinary queries run with remarkable efficiency.

MODULE ASPECTS DONE WELL

One thing that I must give Dr Barnard praise on, is how well structured her lecture slides are. The slides provided me with everything I needed in terms of understanding the concepts. And if you thought that was all, there were videos that were way better than YouTube videos, in quality and the content presented. If I didn't understand something in class, I knew that watching her videos many times would give me my light bulb moment.

Another thing that I liked about this module were the group class activities. Those class activities may have been the only reason I was able to wake up for my 7h45 class. While doing these activities, I was able to make a couple of friends, since it was my mission to sit next to different people every week so I can meet more people. This helped grow my network of like-minded people, and not to mention how fun these activities were.

Finally, another thing that made a difference in my learning this semester was the idea of draft assignments. My partner and I were able to nearly ace all assignments because of the valuable feedback that was given in the draft assignments. The draft feedback gave us a platform to rethink our ideas and how we approached questions which made us perform well in our final assignments.

MODULE ASPECTS THAT NEED IMPROVEMENT

One thing that I found lacking in this module was the absence of student assistants. According to me, student assistants do more than just assist students with course work, but they also share experience as seniors. Getting to know someone who has been on your path is refreshing. It is something that I missed in this module. As amazing as Dr Barnard is, being a lecturer for very long does create some gap between students and lecturers, but someone who did WRDV301 last year may have insights that can help me as a student, such as what to expect in the assessments and so forth.

LIFE LESSON

One lesson I have learned from this module (I guess two in my case), is being detail oriented and always showing up.

In designing a database, a lot of aspects must be taken into consideration. This helped me to sharpen my “attention to detail” skills, which will ultimately help me as I pursue my career in Software Engineering.

In case of showing up, waking up for the 7h45 class taught me resilience which will be useful in the workplace. Due to this, I am now a morning person, implying that everything important should be handled in the morning!